EE 122: Introduction to Control Systems

Problem Set #10: Bode plots for control design

Name: ______________ **Submission Date:** ______________

Problem 1 [40 points] For a second-order transfer function

$$G(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

prove that for frequencies that are much higher than the natural frequency ω_0, the magnitude of the transfer function in decibels is approximately $40 \log \omega_0 - 40 \log \omega$ dB. That is, the magnitude decreases at a rate of 40 dB/decade for frequencies much higher than ω_0. This is the reason why we commonly say that one pole contributes a slope of -20 dB/decade. Here, since there are two poles, the slope is -40 dB/decade.

Problem 2 [60 points] Using MATLAB/Python, plot the Bode magnitude and phase plots for Example 9.13 in the textbook.

- Clearly label the frequencies at which the slopes change (10 points)
- Generate an approximate Bode plot by applying the following logic: a pole at a frequency ω_p contributes a slope of -20 dB/decade for frequencies higher than ω_p, and a zero at a frequency ω_z contributes a slope of +20 dB/decade for frequencies higher than ω_z. Remember that a zero of a transfer function is the value of s that makes the transfer function numerator equal to zero. (30 points)
- Your code should generate same plot as in Figure 9.15 in the textbook. (10 points)
- Write about your approach in words and include your code in the submission. (10 points)

Optional practice problem: Try solving Problem 9.4 — Using a Bode plot for the analysis of a operational amplifier circuit.